Forecasting District-Level Paddy Yield in Sri Lanka with an Explainable Boosting Machine

MSc in AI – Machine Learning Assignment Report

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Source code: https://github.com/WarnaBuddhikaUom/paddy-yield-forecasting-sl

Summary

This project forecasts the average paddy yield (kg/ha) of each Sri Lankan district for the coming Maha or Yala season, using only information available at sowing time. The data are Department of Census and Statistics paddy statistics for 1979–2023. The main model is an **Explainable Boosting Machine (EBM)**, an algorithm not covered in the lectures. EBM is a glass-box generalised additive model with pairwise interactions.

On unseen 2016–2023 seasons the EBM reached a test RMSE of **623 kg/ha**, on par with the black-box tree ensembles (608–633 kg/ha) and **16 % better than the persistence forecast** (22 % excluding the 2022 fertilizer-ban shock). Its exact, built-in explanations agree with SHAP, LIME and partial-dependence analyses, and match agronomic knowledge. A fairness analysis shows the forecasts are least reliable for Northern conflict-affected districts. The model is deployed in a Streamlit web app that explains every prediction.

1. Problem definition and dataset

1.1 Problem and relevance

Rice is Sri Lanka's staple food. It is grown in two seasons: **Maha** (north-east monsoon, about October–March) and **Yala** (south-west monsoon, about April–September). Fertilizer distribution, Paddy Marketing Board purchasing, storage and rice-import decisions must be made **before** the harvest is known, and the 2021/22 chemical fertilizer ban, which cut national yield by about a third in 2022, showed the cost of surprises. The task is to predict, **at sowing time**, each district's average yield for that season. It is a supervised **regression** problem on tabular data.

1.2 Data source, features, target and size

- **Source:** district-wise paddy statistics published by the Department of Census and Statistics (DCS), Sri Lanka, downloaded as a CSV of agricultural statistics (Agriculture_Statistics_Paddy.csv).
- **Raw size:** 2,612 rows × 16 columns. Each row is one district × season × year (1979–2023). The columns are sown and harvested extent (ha) under major irrigation, minor irrigation and rain-fed conditions, average yields and production.
- **Target:** average yield of the district-season, **yield_kg_ha**.
- **Features (all known at sowing):** District, Season, Year; log of total sown area; share of sown area under major, minor and rain-fed conditions; % change in sown area vs the same season last year; yield of the previous season, of the same season last year and the 4-season average; last season's harvested ÷ sown ratio (a crop-loss signal).
- **Excluded to prevent leakage:** the current season's harvested area, per-source yields and production. Production equals net harvest × yield exactly, so including these would leak the answer.
- **Modelling size after cleaning:** 2,174 district-seasons from 25 districts.

1.3 Preprocessing

A data audit found five problems, each fixed using domain knowledge:

- Numbers were stored as text with thousand separators, and "-" marked missing values. These were converted to numbers, with "-" set to missing.
- **100 exact duplicate rows** (2010–2012 were entered twice) were removed.
- **Four regional aggregates** (Mahaweli, Udawalawe, Highland, North & East) overlap real districts and were removed, leaving the **25 administrative districts**.
- **Unit error:** the yield column is labelled "MT/Ha", but its values (1,150–6,600) are **kg/ha**. Production = net harvest × yield / 1000 held exactly (median ratio 1.000), which confirmed the unit.
- Seasons with zero sown area (mostly Northern districts during the war years) were excluded and left as gaps. Lags are joined on a chronological season index, so a gap becomes a missing value rather than silently using an older season.

Encoding and normalisation: the EBM handles categorical strings and missing values natively. For the benchmark models, a scikit-learn Pipeline (fitted on training folds only) applies median imputation, standard scaling and one-hot encoding of District and Season. Sown area is log-transformed because district extents range from hundreds to 100,000+ ha.

1.4 Ethical data use

The data are **aggregated, publicly released official statistics** at district level. They contain no personal, household or farmer-level information and no sensitive attributes, so no consent issues arise. The data and code are shared openly for reproducibility.

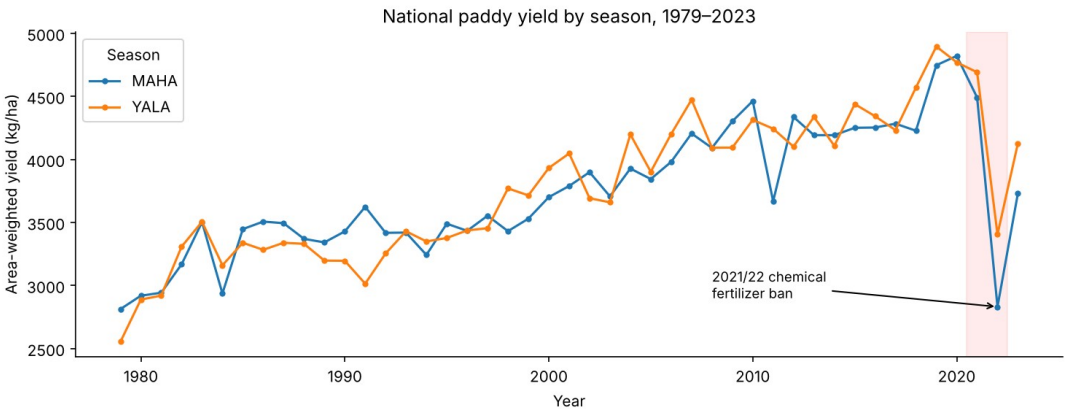


Figure 1. National area-weighted yield by season: yields roughly doubled since 1980, then collapsed in 2022 (fertilizer ban).

2. Selection of a new machine learning algorithm

2.1 The Explainable Boosting Machine

EBM (Lou, Caruana & Gehrke, 2012–2013; implemented in Microsoft's InterpretML) is a **generalised additive model with pairwise interactions (GA^{2}M)**:

$$\hat{y} = \beta_0 + \Sigma_i f_i(x_i) + \Sigma_{(i,j)} f_{ij}(x_i, x_j)$$

Each f_i is a learned **shape function** for one feature, and each f_{ij} is a small 2-D function for one of a few automatically detected feature pairs. Training uses **cyclic gradient boosting**: in each round the algorithm visits the features one at a time and fits a very small tree on that single feature to the current residuals. It uses a tiny learning rate over thousands of rounds, then averages many bagged copies to reduce variance. The result is as flexible as boosting for each feature, yet every effect can be plotted and read.

2.2 Why it was selected

- **Trust in a public-planning setting:** agricultural officers need to see exactly how irrigation mix or last year's yield changes a forecast. In an EBM the shape functions *are* the model, so explanations are exact rather than post-hoc approximations (intrinsic interpretability).

- **Competitive accuracy:** boosting keeps it close to XGBoost and LightGBM on tabular data, which was verified in Section 3.
- **Fits the data:** it handles the categorical District and Season and the missing lag values natively, and its additive structure matches yield, which is largely a baseline level plus an irrigation effect plus a trend.
- **Not deep learning,** and it works well on a small tabular dataset of about 1,800 training rows.

2.3 How it differs from standard models

Model	Form of the model	Interpretability
Linear / logistic regression	One straight-line weight per feature	Transparent, but no curves or thresholds
Decision tree	One tree; any split can mix features	Readable only if shallow; unstable
k-NN	No global function; uses nearest stored examples	Only 'similar cases' explanations
RF / XGBoost / LightGBM / CatBoost	Sum of hundreds of deep multi-feature trees	Black box; needs SHAP/LIME
EBM (this work)	**One learned curve per feature + a few pairwise surfaces**	**Glass box; exact global and local explanations**

3. Model training and evaluation

3.1 Train / validation / test split

- **Training: 1979–2015** (1,782 district-seasons).
- **Validation:** forward-chaining TimeSeriesSplit with 4 folds inside the training years. Each fold trains on the past and validates on the next block of years. All hyperparameters were chosen on this score.
- **Test: 2016–2023** (391 district-seasons, unseen, including the 2022 shock). It was used once, for reporting.
- **Held-out inference case:** Polonnaruwa, Yala 2023, removed from everything and used to demonstrate inference from the saved pickle file.

A random split would let the model learn from future years, so a chronological design was used.

3.2 Hyperparameter choices

Nothing was left at default values. EBM was tuned with RandomizedSearchCV over designed ranges: learning_rate 0.005–0.05 (small steps give smooth curves), interactions {0, 3, 5, 10}, min_samples_leaf 5–30 (each step must be supported by enough district-seasons), max_leaves {2, 3}, outer_bags {8, 14}, smoothing_rounds {0, 100, 200} and max_bins {128, 256}. **Selected:** learning_rate 0.0069, interactions 10, min_samples_leaf 25, max_leaves 3, outer_bags 14, smoothing_rounds 100, max_bins 128. The benchmark ensembles taught in the lectures (Random Forest, XGBoost, LightGBM, CatBoost and a Ridge-stacked combination) were tuned on the same folds, with shallow trees, large minimum leaf sizes and L1/L2 regularisation chosen for the small dataset.

3.3 Performance metrics

RMSE (penalises large misses, which matter most for food-security planning), **MAE** (typical error in kg/ha), **MAPE** (relative error), **R^2**, and the **share of forecasts within ±10 %** (an operational "good enough" rate). RMSE is also reported excluding 2022, to separate normal-year accuracy from the policy shock. The existing approaches used as baselines are **persistence** (same season last year), a **4-season moving average** and **linear regression**.

3.4 Results

Model	Valid. RMSE	Test RMSE	MAE	MAPE %	R^2	±10 %	RMSE ex-2022
Stacking	430	608	443	12.5	0.54	58	451
CatBoost	437	608	450	12.7	0.54	58	465
Random Forest	430	622	463	12.9	0.52	56	485
EBM – new algorithm	**441**	**623**	**449**	**12.9**	**0.52**	**60**	**457**
LightGBM	425	626	451	12.9	0.51	60	455
XGBoost	431	633	457	13.0	0.50	59	465
Linear regression	513	639	467	13.6	0.49	58	461
Moving average (last 4)	442	685	496	14.2	0.42	55	517
Persistence (same season LY)	477	742	554	15.5	0.32	50	583

Table 2. Errors in kg/ha. Validation = forward-chaining folds on 1979–2015; test = unseen 2016–2023. Sorted by test RMSE.

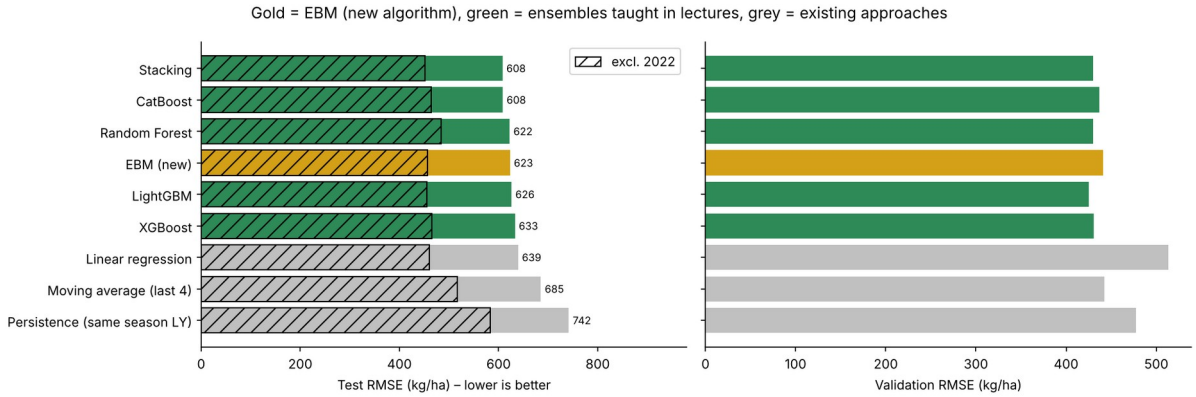


Figure 2. Test and validation RMSE. Gold = EBM, green = ensembles taught in lectures, grey = existing approaches.

What the results indicate

- **On validation,** the tree ensembles score 425–437 kg/ha and EBM 441, about 4 % behind the best (LightGBM). EBM is still 14 % better than linear regression and 8 % better than persistence.
- **On the unseen test years,** EBM's RMSE of 623 kg/ha beats LightGBM (626), XGBoost (633) and linear regression (639). It is **9 % better than the moving average and 16 % better than persistence**, and within 2.5 % of the best model (stacking, 608). Its MAE is the second best, and 60 % of its forecasts fall within ±10 %.
- **Interpretability is therefore almost free:** a fully transparent model matches black-box accuracy on future seasons.
- **2022 is the exception:** mean errors reach about 1,400 kg/ha in Maha and 900 kg/ha in Yala, against 200–540 kg/ha in other seasons. Every model over-predicts that year.
- **On the unseen Polonnaruwa Yala 2023 case,** the pickled model predicts **4,731 kg/ha** against an actual 4,918 kg/ha (−3.8 %).

4. Explainability and interpretation

Four methods were applied: (1) EBM's intrinsic global and local explanations; (2) Partial Dependence and ICE plots; (3) SHAP on the best black-box benchmark (LightGBM) as an independent cross-check; and (4) LIME for the held-out case.

4.1 Which features are most influential

The largest EBM terms (mean absolute contribution) are: same-season yield last year (≈162 kg/ha), Year (≈158), rain-fed share (≈138), District (≈113), 4-season average yield (≈104), major-irrigation share (≈91)

and Season (≈73). All pairwise interactions are small (≤ 20 kg/ha), so district yield is almost additive, which is why a GAM can match boosted trees here.

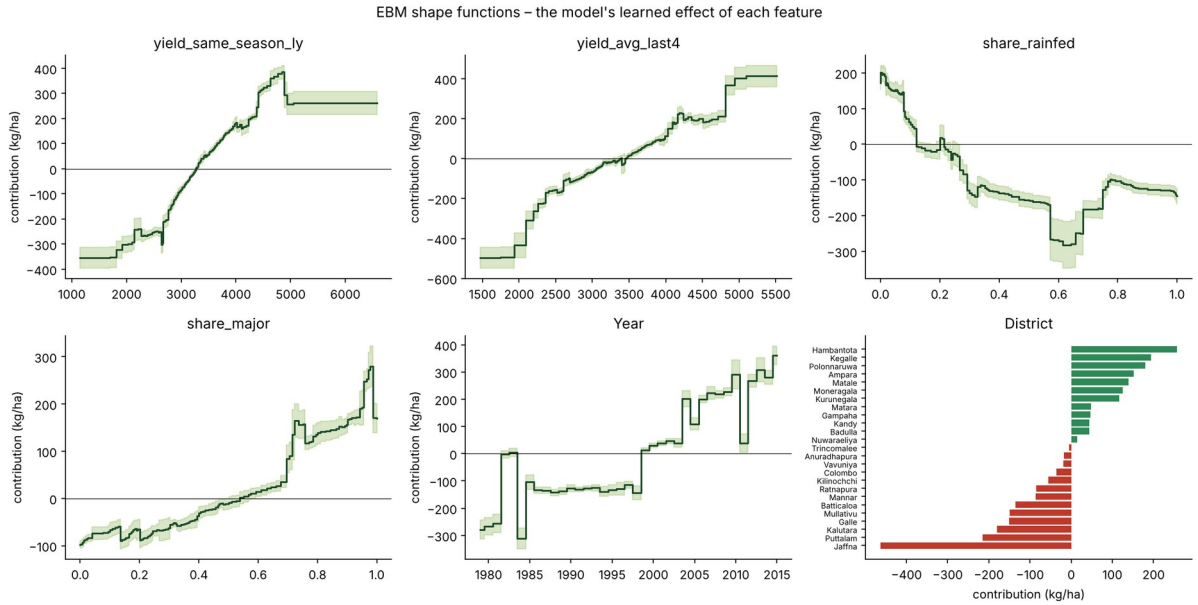


Figure 3. EBM shape functions: the model's learned effect of each feature (bands show variation across bagged models).

4.2 What the model has learned

- **Yield history:** contributions rise almost linearly with past yields, so good districts tend to stay good (persistence).
- **Rain-fed share:** a clear downward curve, from about +200 kg/ha (no rain-fed area) to about −150 to −250 kg/ha (mostly rain-fed). That is a gap of roughly 400 kg/ha caused purely by water security.
- **Major-irrigation share:** flat until about 60 %, then a sharp rise of up to +250 kg/ha for districts dominated by large reservoir schemes.
- **Year:** the technology trend appears as steps, low in the 1980s–90s and rising through the 2000s and 2010s.
- **District:** once history and irrigation are known, district terms are adjustments. They are positive for dry-zone irrigated districts (Hambantota, Polonnaruwa, Ampara) and negative for Jaffna and some wet-zone districts. **Season:** Yala is lower than Maha.

4.3 Cross-checks with PDP, SHAP and LIME

The PDP/ICE curves show the same directions as the shape functions. SHAP on LightGBM, a different model explained post hoc, puts the same features at the top: history, Year, rain-fed share and major share. The one disagreement is District (4th for EBM, 12th for LightGBM-SHAP). LightGBM spreads district identity over 25 one-hot columns and lets the correlated history features carry the location effect, while EBM gives District its own term.

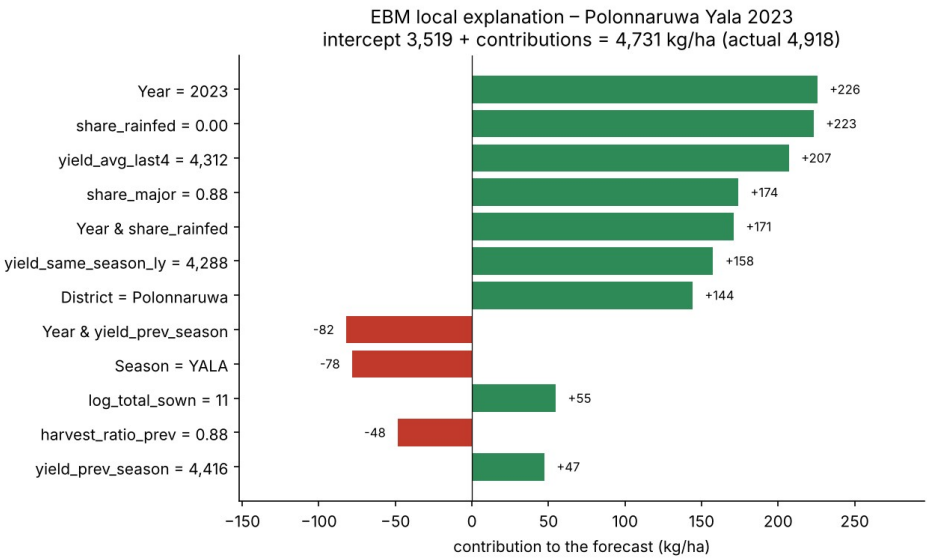


Figure 4. Exact local explanation for the unseen case, Polonnaruwa Yala 2023.

Local explanation: 3,519 kg/ha (average) + recent Year (+226) + no rain-fed area (+223) + high 4-season average (+207) + 88 % major irrigation (+174) + Year × rain-fed interaction (+171) + last year's yield (+158) + district (+144) − Yala (−78) + small terms = **4,731 kg/ha**. LIME (local surrogate $R^2 \approx 0.5$) lists the same drivers with the same signs, but only approximately; the EBM breakdown is exact.

4.4 Alignment with domain knowledge

Water security (major irrigation raises yield, rain-fed lowers it), persistence of district performance, the long-run technology trend and a weaker Yala season are all well-established features of Sri Lankan paddy production. No leakage variable or suspicious feature dominates. The model behaves the way an agronomist would expect, which is the basis for trusting it.

5. Critical discussion

5.1 Limitations of the model

- **No extrapolation:** EBM's curves, like trees, are piecewise constant, so record yields beyond the training range (2019–2021) are under-predicted. The deployed model is retrained on data up to 2023 and must be retrained every season.
- **Shocks are invisible:** the fertilizer ban, floods and droughts after sowing are not in the features. Errors in 2022 were 2.5–4 times normal.
- **District averages only,** with a point forecast and a typical-error band rather than a full probabilistic forecast.
- **Small data:** about 2,200 district-seasons, with validation scores varying by 40–60 kg/ha between folds. Differences of 1–2 % between models are not meaningful.

5.2 Data-quality issues

Duplicates, overlapping aggregates, the unit label error, text-formatted numbers and placeholders were found and fixed (Section 1.3). The war-year gaps in the North remain, handled as missing values. The DCS figures are official estimates built from field surveys and administrative reporting, so they carry measurement error. District boundaries also changed over 45 years (for example, Kilinochchi district was created from Jaffna in 1984).

5.3 Risks of bias or unfairness

Errors were broken down by district group on the test seasons, excluding 2022:

District group	Districts	MAPE %	Mean bias (kg/ha)	Within ±10 %
Mixed	7	8.8	-43.2	68.4 %
Mostly irrigated	7	6.6	-45.8	82.5 %
Mostly rain-fed	6	8.8	+60.6	61.9 %
Northern, conflict-affected	5	12.4	-131.0	54.0 %

Table 3. EBM test errors by district group (excluding 2022). Positive bias = over-prediction.

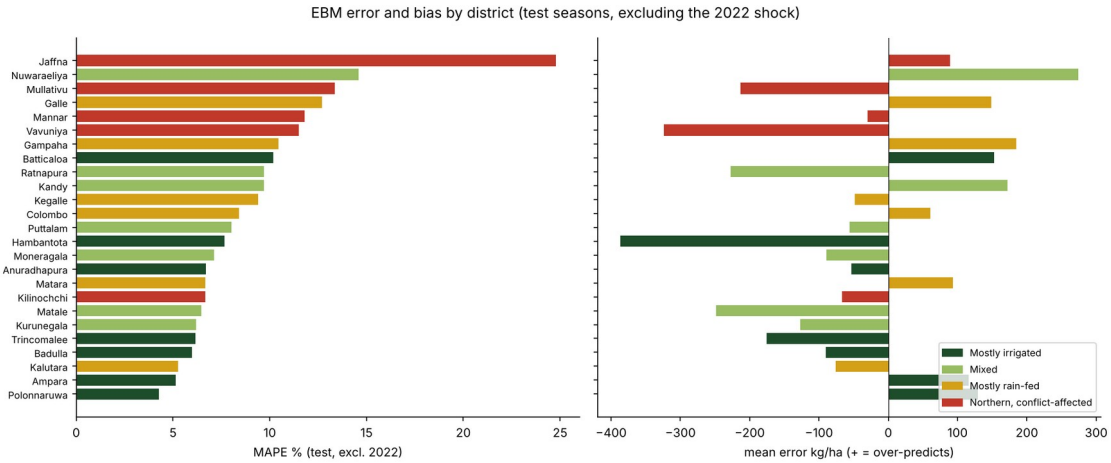


Figure 5. EBM error and bias by district.

Forecasts are most reliable for irrigated districts. **Northern conflict-affected districts** have nearly double the error and are **under-predicted**, because their history has gaps and their post-war recovery is a trend the model has seen little of. **Rain-fed districts** are **over-predicted**, because rainfall is not a feature. If forecasts were used to allocate fertilizer, credit or procurement quotas, these errors could systematically disadvantage those regions. Mitigations: show district-specific error bands, keep a planner in the loop, add rainfall and reservoir data, and monitor per-district error each season. The data contain no individual-level attributes, so the fairness concern is between regions.

5.4 Real-world impact and ethical considerations

- **Benefit:** early, explainable district forecasts can help plan fertilizer, storage, purchasing and imports, and can flag districts likely to fall short.
- **Human oversight and accountability:** the model supports planners and does not replace them. Exact explanations let officers see *why* a district is predicted low and challenge it with local knowledge.
- **Over-reliance:** a precise-looking number can hide a shock the model cannot see, as in 2022. The app shows an error range and states its limits.
- **Data ethics:** the data are public, aggregated and non-personal, and the code is open for scrutiny.

6. Front-end integration (bonus)

The final EBM and its metadata are saved as **paddy_yield_model.pkl**. A **Streamlit** web app, generated with GenAI prompts that are documented in the repository, loads the pickle and pre-fills a district's history from the DCS records. Users can enter or edit the sown area by irrigation type and the recent yields. The app shows the predicted yield, a typical error range and an indicative production, plus the EBM's **exact explanation chart** for that prediction. A Dockerfile and a GitHub Actions workflow (tests on the pickle, Docker build) support deployment. A 3–5 minute demo video accompanies this report.

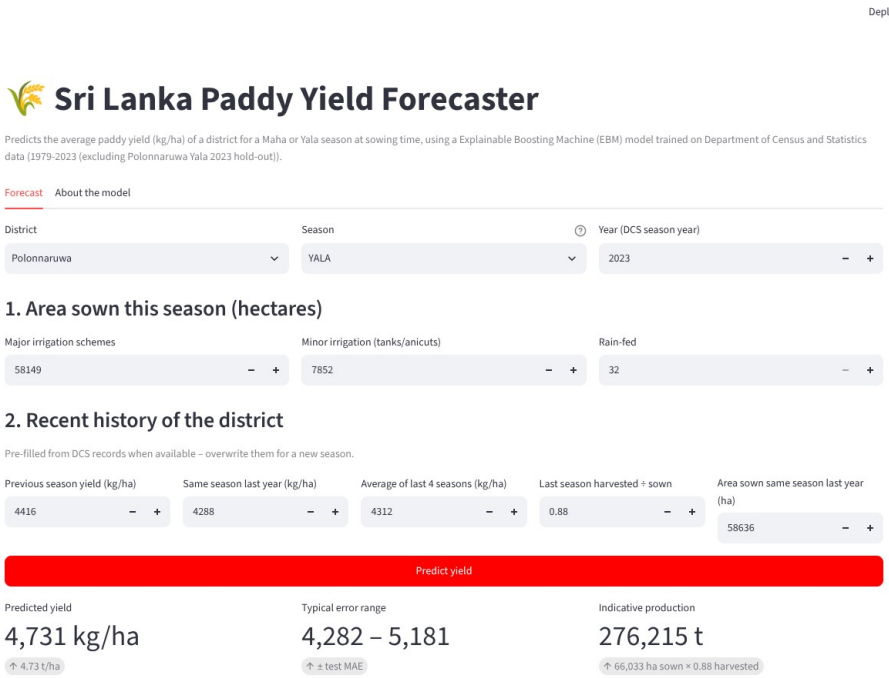


Figure 6. The Streamlit front end: inputs and prediction.

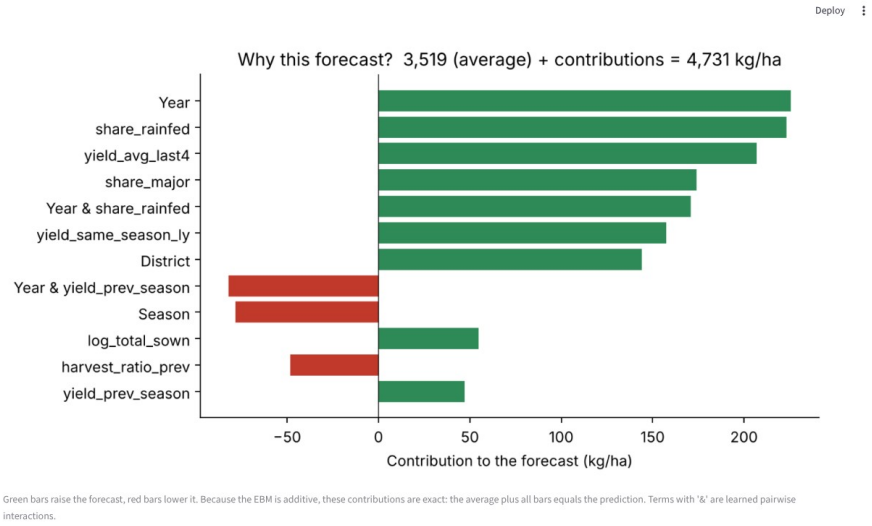


Figure 7. The app's per-prediction EBM explanation.

7. Conclusion and future work

A local Sri Lankan problem was solved end to end: careful cleaning and leakage-free features, a new glass-box algorithm (EBM) with designed hyperparameters, time-aware evaluation, and four explainability methods. The EBM matches black-box accuracy on unseen seasons, explains every prediction exactly, and behaves consistently with agronomy. Its limits, namely no extrapolation, blindness to shocks, and weaker accuracy for Northern and rain-fed districts, are documented. **Future work:** add rainfall, reservoir and fertilizer-issue data; produce prediction intervals; monitor per-district error and data drift with automatic retraining (MLOps).

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